Appendix

FORM 1



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Programme:RSWS3Y2 (Group:G14)

Assignment

BMSE 2013 SOFTWARE QUALITY AND MEASUREMENT

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Date of Submission: 4/17/2026

FORM 2

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

Plagiarism Statement and Guideline for Late Submission of Coursework Read, complete, and sign this statement to be submitted with the written report. **We confirm that the submitted works are all our own work and are in our own words.**

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BMSE2013 - Group Assignment Rubrics (CLO3)

Student Names:	(1)	Tutorial Group:		/80
	(2)	Programme:	R ( )	Report presenation /20
	(3)			
	(4)			
	(5)			

Section	Criteria / Area	Excellent	Good	Average	Poor	Very Poor	Score
PART 1	• **Introduction to system**	Very detail elaboration on introducing background of the chosen organization. (5m+5m)	Good elaboration on introducing background of the chosen organization. (4m+4m)	Generally show attempt on introducing background of the chosen organization with minimum findings (3m+3m)	some important information is missing in the elaboration.. (2m+2m)	very poor findings of organisation background (1m+1m)	
	• **Quality attribute selection**	Very good elaboration on factors with mximum justifications (13m-16m)	Good elaboration on factors with justifications but it can be improved further (9m-12m)	Generally show attempt on providing elaboration on factors with 3 justification (5m-8m)	provide elaboration on factors with 2 justification (3m-4m)	provide elaboration on factors with 1 justification. (1-2m)	

	• **Software Quality management process**		Very good elaboration on quality management process with sufficient findings of the explaination	(7m-8m)+1 mark	Good elaboration on the quality management process with but findings which can be improved	(5m-6m)+1 amrk	Generally find the quality management process with limited elobration and findings	(3m-4m) +1 mark	provide minimum ellobartion which need to be improved on quality management process and need to more findings	(2m)+1 mark	very little ellobration of the quality management process and need a major enhancement	(1m)+1 mark					
	Lecturer comments																
	Total 35																

	Criteria / Area	Excellent		Good		Average		Poor		Very Poor		Score				
												(1)	(2)	(3)	(4)	(5)
part 2	**software Quality metrics**	two relevant tool are identified and application of each tools enhances system quality with excellent findings	(13m-15m)	Good identifiation in three relevent matrix but there are some minor changes to be improved in application of each matrix further.	(9m-12m)	very average findings of revelent matrix and application of matrix need to more detailed by finding more research on the application	(5m-8m)	Inadequate attempt to provide the revelent matrix and need research in the quality and the application	(3m-4m)	Very little attempt on the finding the matrix and application is incorrect and need major correction on the findings .	(1m-2m)					
	Quality Control Measurements Tools	Very good findin g on the software measurement tool which able to relate to the given projec by providing excellent justification	(20m)+10 marks	Good discussion on software measurement tool and discuss with relevent finding yet still need some minor correction	(16m)+8marks	very average discussion of the software measurement tool which need some improvement and relate this with some relevent project	(11m)+6marks	Little attempt in discuss the software measurement tool which failed to relate how to apply the project with poor justification	(5m)+3marks	Very poor attempt in discussion on the software measurement tool and failed to discuss the project and NO justification	(1m) +0 mark					

	Lecturer comments																	
TOTAL (45 marks)																		

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1. Introduction

Taobao was launched by Alibaba Group in 2003, now it is one of the largest global platforms with over 500 million users and offers a variety of product categories such as fashion, electronics, home products, and handmade products. It first started as a Consumer to Consumer (C2C) marketplace but slowly evolved into a hybrid model that combines both C2C and B2C (Business to Consumer). Recently in the year 2024, Taobao added an English Language interface, this change allowed Taobao to expand its wings internationally resulting in a large boost of international user growth.

Taobao aids in connecting individual sellers, small business, and large corporations with global buyers. The system provides all the necessary functions to allow both buyers and sellers to use it with ease. These functions include integrated payment systems to make payments, logistics coordination to track the purchase of goods, marketing tools to increase brand visibility, and data analytics to allow sellers to make informed decisions based on the analytics. Taobao's earns their revenue through 3 methods, the first is advertisement where sellers pay the platform to advertise their products on the 'sponsored search' slots so the products are seen by buyers, second the transaction fees where the sellers pay a small transaction fee of 0.6% for payment processing through Alipay while larger brands pay a higher commission fee of 2% to 5% and an annual server fees to keep their premium status, and lastly premium seller features which are paid subscriptions that sellers purchase to gain a more advanced data analytics that aids sellers run their businesses more strategically.

There are various stakeholders involved with Taobao. The first stakeholders are buyers who are the consumers of the platform where they seek products through the application, buy a product of their choice, and receive the goods through the reliable delivery service provided. Sellers contain a diverse

amount of groups that range from individual sellers to large scale merchants. The Alibaba Group, they guide Taobao as it is their subsidiary in a direction after making strategic decisions and providing the necessary resources for it to run. Operational partners which includes Alipay, which is an integrated payment processing system for Taobao and Cainiao Network which manages logistics coordination. Other stakeholders include third party delivery firms where they deliver the purchased goods to its buyer correctly, the customer service team where they assist both buyers and sellers for any inquiries, and the government regulators which enforce rules that protect the consumer rights and ensure Taobao operates accordingly.

Taobao has encountered multiple incidents that highlight the importance of software quality during the entirety of its operation. The first incident happened between the years 2010 and 2013 where the system experienced a downtime during an important sales event called the “11.11 Singles Day” where the traffic spike overwhelmed the system capacity leading to millions of buyers unable to complete their orders. Another incident happened in 2015 where a data breach occurred and compromised the merchants credentials which led to fraudulent transactions and exposed weak security controls. In 2020, a logistics failure which occurred during the pandemic where the rise in demand of buyers staying inside unable to go out caused the rise and the logistic partners to struggle with the order consistency, this resulted in the delivery being delayed and inventory errors occurring often.

In summary, Taobao’s journey as a C2C startup in 2003 to a global e-commerce company with billions of users shows that it has grown significantly throughout its operation. To transition into a hybrid business model with support from a wide range of stakeholders has helped in creating a highly dynamic marketplace. However, past incidents indicate that even such a large platform is not without fault. These events highlight that Taobao has continuously proven its success where it relies not just on its scale, but on maintaining their strong approach to managing their quality and ability to learn from failures.

1.1 Main Features

Taobao's modules all work together to make the platform easy to use. To identify if the software is of high quality, we will first look at the main features. These features are critical because they need to be fast, accurate, and secured. This section will explain each of the features:

1.1.1 Product Search

This feature function is by allowing the user to find the items they want, by simply typing into the search bar provided to find what they want from the millions of items in Taobao's database, the application will then scan every product to display the relevant result that matches what the user is searching for. There are buttons provided for users to filter their search results such as prices, brands, items near the buyer's location, this function allows users to narrow their search results down and a sorting function is also provided where you can list the order of items by most or least popular items first. If a user makes a typo in the search bar an autocorrect is provided to fix your mistakes and as you type, the app will show you the related search terms.

1.1.2 Recommendation

This feature displays items customized to the buyers preferences. It gets information through what the buyers clicked on before, what the buyer bought, and the search history. The system will study information to understand the buyer's preferences and identify the items they might like. The recommended products will then be displayed on the buyers homepage and even under the product page. This feature allows Taobao to sell more of its products by displaying buyers interests that they have not actively searched for.

1.1.3 Payment

This module handles all the transactions occurring on the platform. Everything purchased goes through Alipay, which is Taobao's built-in partner for transactions. It works by the buyer clicking "buy", the app then checks if the money is sufficient and starts the transfer. There are multiple ways to pay this including credit or debit card, e-wallet, or even installments. This module uses an Escrow method where

the seller does not get the money instantly, the platform will hold onto the money until buyers make confirmation on getting their package safely.

1.1.4 Order Tracking

This module allows buyers to track their order after payment is confirmed. The application will communicate with logistics partners such as Cainiao and other delivery companies to display the live location of the purchased product. It will display the whole journey starting from when the seller packs the product, to when the product is on the truck, and finally to when it is arriving at the destination provided. The module will also display an estimation of when the product will be delivered to give the users an idea of when it will arrive. There is also a return tracking where if the item purchased is subjected to return by the buyer when it is not to their satisfaction, the return package can also be tracked when going back to the seller to ensure it is received.

1.1.5 Review and Rating

After purchasing an item and receiving it, buyers are allowed to write a review and rate the item by giving it a score from 0 to 5 stars on how satisfied the buyer is with the product. These reviews will then be displayed under the product page so that other buyers who are interested in it can make decisions if the item is worth purchasing. The system will then calculate the average score for the product and display it under the product page. This module also provides a security feature where it detects fake reviews and flags them for moderators to review and take action. By allowing buyers to share about their experience with the product and leaving a review helps build trust within the marketplace.

Conclusion

All these features mentioned are the backbone of Taobao's online shopping platform. By understanding and checking these features we can identify whether the app is reliable. Ensuring these features work as intended is how Taobao stays successful.

Summary Table

Feature/Module	Description
Product Search	● Allows users to find products by typing into the search bar. ● Scans millions of listings to return relevant results. ● Filters for price, brand, seller location, and popularity. ● Includes autocorrect for misspellings and related term suggestions.
Recommendation	● Generates personalized product suggestions for each user. ● Analyzes browsing history, past purchases, and search patterns. ● Displays recommendations on homepage and product pages. ● Helps users discover items they may not have actively searched for.
Payment	● Handles all transactions through integration with Alipay. ● Processes transactions, verifies funds, and manages payment transfers. ● Supports multiple payment methods like credit cards, e-wallets, and installments. ● Manages escrow, holding funds until the buyer confirms receipt of goods.
Order Tracking	● Provides visibility into order status after purchase. ● Integrates with Cainiao and third-party logistics for real-time data. ● Updates users on shipment status (shipped, in transit, out for delivery). ● Supports tracking for returns when items are sent back.
Review and Rating	● Allows users to submit text reviews and a 0 to 5 star rating. ● Displays feedback on product pages for potential buyers. ● Calculates the overall product score from users ratings. ● Flags any potential fraudulent reviews for moderators.

1.2 Critical Risk if Quality Fail

Risk management is a process of identifying, checking and managing threats to a company's money and operations. The role of software quality is to shield against threats that could cause the quality to fail, if poor quality it will lead to various risks ranging from internal coding bugs to external partner failures which will harm Taobao;s reputation.

1. Product Search Module

- **Technical Risk:** If search tool quality is low the cross language search algorithm will contain coding errors for processing the English keyword into Chinese characters to search for the item that is named in the Chinese product listing specifically. This can impact the performance or functionality of the feature where the translation bug will make it not perform as intended for international users.

- **Business Risk:** If search displays wrong items to buyers, Taobao fails its main goal of connecting buyers to the right products. Users will be unable to find the product they are looking for and will then abandon the platform, this results in loss of sales and revenue.

- **Operational Risk:** If search is implemented poorly when integrating with multilingual search then it will lead to confusing filters or unclear navigation, international users will have a hard time looking for products they want which leads to an increase in customer support tickets for inquiries, and operational costs due to the decrease of systems efficiency.

2. Recommendation Module

- **Business Risk:** If the system is of poor quality and provides the wrong features for example displaying items that are not in stock or already bought will result in users not being given anything of their interest or display products that are helpful in driving them to purchase products on the platform. Taobao will lose the chance to make any “unplanned” sales happen to gain extra profit and sellers who paid for advertisements to display their products as a recommendation for

users will get no results, leading to a loss in a return of investment (ROI) for sellers and damaging sellers trust towards the platform.

- **Technical Risk:** Recommendation algorithm has been trained on Chinese data for years which will then contain flawed logic when applied to international users such as western fashion, sizing, or seasonal differences. There will be a performance risk where the software does not function as intended and produces outputs due to algorithms not being redesigned properly for cross-cultural platforms.

- **Project Risk:** If quality is low due to not researching properly about the international market to produce quality recommendation features (skipped Requirement Validation Phase) then the quality will suffer. The product management team decisions will lack a cultural Subject Matter Expert (SME) input which can increase the risk of Functional not performing as its intended purpose in the recommendation engine.

3. Payment Module

- **Technical Risk:** If the software quality is low and causes bugs between the Taobao and Alipay functionality such as the sending the wrong bill to the customer it will make important features like the escrow system fail. Payment is the most critical module if this fails and not function as intended

- **Security Risk:** There are different data protection laws for different countries such as General Data Protection Regulation (GDPR) is used in Europe and California Consumer Privacy Act (CCPA) is used in California and the system needs to adapt correctly to each standard to ensure data is protected accordingly. Failure to implement the localized data encryption and access control protocols for each can lead to non-compliance which can result in financial penalties, loss of licenses to operate in the market, and a breach of users Personal Identifiable Information.

- **Operational Risk:** If the payment system experiences downtime during peak hours especially for international buyers due to poor load handling, users will be unable to complete their purchases. This will disrupt the daily operations of Taobao and cause an immediate revenue loss.

4. Order Tracking Module

- **Technical Risk:** The module must connect to many different couriers APIs such as FedEx, DHL, and more. These all have different standards to follow which can possibly lead to tracking errors if the system does not communicate correctly with each other, especially in countries with poor digital infrastructure. This can lead to increase of customer support tickets and loss of users trust if API fails.

- **Operational Risk:** Handling returns is problematic because each country has different shipping labels and costs. If the app does not provide a clear local return address, the user might get stuck with the product they do not want. If Taobao's inventory is not synchronized and is not 100% accurate it can lead to a refund loop because on the app it says in stock but in the warehouse there is no stock. These can lead to shipping delays and unsatisfied customers.

- **Programmatic Risk:** If external delivery partners experience any system outages or technical failures. The software must implement an error handling that can provide informative and helpful messages to the users affected to reduce their worry over the product they have purchased, if this is not implemented and users just see a blank screen this confusion will increase customer support tickets and user frustration.

5. Review and Rating Feature

- **Business Risk:** If the software quality is low to detect “bot” patterns or fake accounts under the products review and rating page it will result in fraudulent reviews following the products review and rating section. This will destroy the integrity and honesty of the seller while will lead to

Taobao failing in its objective of being a trustworthy platform, and buyers will find it hard to trust in the reviews and ratings which will impact the sales negatively.

- **Technical Risk:** A bug in the code that calculates or displays data is wrong which will then result in the system showing the wrong average rating of the product. For example, showing 2 stars when the product has 5 star reviews overall. This will be a failure in functionality of the product and users will be unable to make any informed decisions on whether to purchase or not due to the failure in calculation. Other than that, a fake review detection algorithm may have flawed logic when analyzing English text. This can cause false positives where actual English reviews are flagged as fake, and false negatives where fake reviews in English are missed. The integrity of the review system will be compromised then.

Summary Table

Feature / Module	**Critical Risk Type**	**Description**
Product Search	Technical Risk	Cross language translation bugs cause English words to mismatch Chinese product listings, impacting the functionality for international users.
	Business Risk	Wrong search results after integrating cross language translation prevent users from finding products which leads to loss in international sales and revenue.
	Operational Risk	Poor multilingual implementation increases

		customer support costs and decreases operational efficiency
Recommendation	Business Risk	Providing "wrong features" (out-of-stock items) results in lost "unplanned" sales and a poor ROI for advertisers.
	Technical Risk	Algorithm trained on Chinese data applies flawed logic to international users (fashion, sizing, seasons)
	Project Risk	Skipped requirement validation and lack of cultural SME input cause recommendation engine to fail its intended purpose
Payment	Technical Risk	Functional bugs between Taobao and Alipay cause billing errors and escrow failures, breaking the core purchase journey.
	Security Risk	Non-compliance with international data laws (GDPR, CCPA) leads to financial penalties and data breaches
	Operational Risk	System downtime during peak international hours disrupts operations and causes immediate revenue loss

Order Tracking	Technical Risk	Multiple courier APIs with different standards cause tracking errors and loss of user trust
	Operational Risk	Inconsistent return processes and inaccurate inventory synchronization cause shipping delays
	Programmatic Risk	Poor error handling during external partner outages shows blank screens instead of helpful messages, increasing user frustration
Review & Rating	Business Risk	Inability to detect "bot" patterns allows fake reviews, destroying marketplace integrity and the platform's trustworthy reputation.
	Technical Risk	Coding bugs lead to incorrect average ratings (e.g., 2 stars instead of 5), preventing users from making informed decisions.Fake review detection algorithm fails for English text, causing false positives and false negatives, compromising review system integrity.

1.3 Quality Attributes selection

To address the critical risk identified, five quality attributes have been selected and ranked according to their importance in mitigating any potential failures. Each attribute is mapped to specific features and risks, with ranking based on most critical attributes to least and justified based on the potential cost impact of failures towards the platform.

1. Integrity

Integrity is selected as the most critical quality attribute for Taobao because it addresses the Security Risks and Technical Risks in Payment Module identified in section 1.3. According to the McCall's Factor model, this factor deals with the software systems security, that is, to prevent access to any unauthorized access. For Taobao it means ensuring that only authorized users can access payment functions and that users' sensitive information will be protected from any unauthorized access.

The risks addressed are the payment modules technical risk where the bugs in Taobao and Alipay integration can cause billing errors to be sent and escrow system to failure, security risks where if Taobao is not complying with the international data protection laws for each country to follow such as GDPR and CCPA there will be a possibility of Taobao facing legal consequences and loss of licenses to operate in the marketplace. Other than that, data breaches that could expose users to sensitive information. To achieve high integrity, Taobao must implement a localized data encryption following each region's standards and establish strict access controls based on users roles for payment functions. There should be regular testing to be conducted such as penetration testing to identify any security vulnerabilities before it can be exploited by hackers. Security audits should be performed to ensure the platform's compliance with international data protection laws and implementation of a good authentication mechanism for all payment transactions that happen in the platform.

Integrity is ranked first because payment and data security failures has the most severe consequences if they occur. If integrity were to be compromised, Taobao will receive fines from GDPR or CCPA due to platform violating regulations which can reach millions of dollars. If Taobao fails to comply with the

operating licenses requirements for each country, there is a possibility of Taobao being banned from the country's marketplace, which can cause financial revenue loss and loss of opportunity to grow. Other than that, if data breaches were eto occur and users information is exposed and used maliciously such as performing fraudulent transactions it can permanently damage users trust in the platform having an increase of users switching to competitor platform and reputation damage will take years to recover if data is breached.

From a cost perspective, it is cheaper to implement penetration testing and security audits compared to the cost of recovery where it includes legal settlements, customer compensation if data is compromised, regulatory fines due to non-compliance, irreparable reputational damage, and lost of revenue from market bans that far exceeds the investment that could be put into compliance measures.

2. Reliability

Reliability is selected as the second most important quality attribute for Taobao because it addresses the Operational Risk in Payment Module, and Technical and Programmatic Risk in Order Tracking Module. According to the McCall's Factor model, Reliability focuses on minimizing the failure rate of the software to ensure that the system performs consistently under stress. For Taobao, it means the system must remain stable during peak events and handle external partners' failures appropriately.

The risk addressed includes payment modules operational risk where system downtime during peak hours causes immediate revenue loss due to the buyers purchasing failing which will make them abandon it. Order tracking modules technical risk addressed is Taobao’s multiple courier services APIs with different standards can cause tracking errors which leads to an increase in customer support tickets and reduction in users trust, and programmatic risk that is platforms poor error handling due to external partners failures which then shows a blank screen instead of informative messages that are helpful in these situation will lead to an increase in users frustration. To achieve high reliability, Taobao should implement load testing to simulate traffic to ensure the system remains stable during peak events. Establish the necessary failovers system where backup servers take over instantly if the main system were to fail. Design a good and appropriate error handling that displays informative messages during external partner downtime instead of blank screens. Monitor the APIs connected to external courier services and alert the teams beforehand to prepare accordingly when failure occurs.

Reliability is ranked second due to it causing great financial loss if the system were to experience a downtime that cannot be recovered. And if API failures were to occur it would increase the customer support team's labor cost and users who experience any tracking failures will lose their trust towards the platform and may switch to the competitor platform, if poor error handling during outages can create uncertainty and anxiety if buyers are not assured about their purchases.

From a cost perspective, the cost of implementing preventions is cheaper such as load testing and failover systems to provide back is lower than the loss in revenue from downtimes, high reliability ensures that the increase of customer support tickets will reduce the operational labor cost. Comparing prevention to recovery, recovery will cost more as the revenue will be affected significantly and compensation must be made to customers after system crashes such as coupons and discounts due to the significant number of users the cost is likely very high.

3. Correctness

Correctness is selected as the third most important quality attribute for Taobao. It will address most of the risk across Product Search Module, Recommendation Module, Order Tracking Module, and Review and Rating Module identified in section 1.3. According to McCall’s Factor Model, correctness ensures the accuracy and correctness of output, it makes sure that the data is displayed as expected. For Taobao, this is important to ensure that search results must match what users are looking for, recommendations must align with users preferences, inventory is accurate, ratings are calculated correctly and fake reviews are flagged accurately.

The risks that will be addressed in this quality attribute are product search modules technical risks, where cross language bugs can occur and cause mismatch between the Chinese product listings, and business risk where search results are inaccurate which causes users to abandon the platform which leads to loss of sales and revenue. In the recommendation module the business risk is addressed where wrong recommendations can cause unplanned sales and advertisements that sellers purchase to be recommended to users will lose their ROI which results in damaging the sellers trust towards the platform and loss of sales, technical risk where the algorithm which was originally trained in Chinese data will have flawed logic when transitioning to international users due to the difference in fashion, sizing, and seasonal differences, and project risk where the skipped requirement validation phase and lack of cultural SME can cause the recommendation to fail its intended purpose due to lack of research.

Furthermore, the review and rating modules risk addressed are business risk where fake reviews often are not detected which then destroys the platforms integrity and credibility, technical risk where the calculation bugs displays incorrect average rating for sellers preventing informed purchasing decisions made from buyers, and fake reviews detection algorithm fails for english text which causes false positives and false negatives will then compromise the review systems integrity which also also destroy users trust in this module.

To achieve high correctness, It is best to implement a unit testing for rating calculations to ensure that calculations are accurate, conduct an integration testing between Taobao and Alipay links to ensure a smooth transaction and that data remains consistent, and test the recommendation logic to ensure that it matches the users preferences and behaviours. Furthermore, by implementing a logic based checks that allows the platform to identify both reviews and fraudulent accounts accurately is important to ensure the rating and reviews credibility, and include a SME in requirements validation phase to ensure that international requirements are met and researched well to ensure a smooth transition internationally. Lastly, test the translation layer thoroughly with various and diverse english search terms, perform retraining of an algorithm that includes international users data, and conduct a regular audit to ensure inventory is accurate for synchronization.

If correctness were to fail, the accuracy failures can slowly destroy the users trust in the system over time which is sometimes difficult to detect, and users who sees wrong search results may just simply leave the platform without purchasing which results in loss of sales that could impact the platform greatly if not detected in time. Other than that, failures in recommendation affects the sellers ROI and will make them migrate to other competitor platforms which then reduces Taobao advertising revenue, if fake reviews not flagged and action taken accordingly it will destroy the marketplace integrity which makes it difficult for buyers to make informed decisions based on the review and rating and inaccurate calculations toward the rating can reduce users confidence in the platform. Lastly, recommendation failures can occur and cause missed opportunities of unplanned sales which help earn extra profit will be loss.

From a cost perspective, investing in algorithm testing is cheaper than losing advertisement revenue due to poor recommendation targets. If correctness fails it will slowly destroy the reputation of Taobao and regaining the users back to the platform will have extensive efforts that will cost a lot. By implementing the appropriate prevention strategies for each risk identified it will still be significantly cheaper compared to recovering the users lost, reputation damage, and the loss of revenues that could be earned if correctness do not fail.

4. Usability

Usability is selected as the fourth most important for Taobao. It will address the Operational within the Product Search module. According to McCall's Factor Model, Usability deals with resources needed to operate the system and ensure users can navigate through the system efficiently with little to no confusion. For Taobao, it allows users to easily search, filter, and fine products without needing to contact customer support for assistance.

The risk addressed in the product search module is operational risk where poor integration of multilingual navigation and unclear filters can cause users to struggle with navigating through the platform for finding products which then leads to an increase in labor cost due to the inquiries sent to customer support teams.

To achieve high usability, conducting a user testing with real international customers can allow us to identify any confusing elements when launching, creating a clear filter and sorting options for product search that are intuitive for users and providing autocorrect for any misspelled words in the search bar correctly according to selected language. By maintaining a consistent navigational pattern across all platforms allows for easy navigation when switching between languages and designing clear error messages helps guide users instead of making them confused.

Usability is ranked fourth and the lowest in the rankings because improvements only involves adjusting the interface rather than performing major changes to the system. Usability failures can cause user frustration but will rarely result in business failure, poor usability leads to increased labor cost but does not cause significant revenue loss as the issues can be addressed through an iterative process of user testing and interface updates.

From a cost perspective, usability only involves improving the interface by making adjustments which are less expensive than major changes the other quality attributes bring. Cost of poor usability will only have at most an increase in customer support team labor cost from confused customers unable to navigate through the app. Furthermore, loss of sales still happen from users who abandon the system due to poor usability and it is difficult to measure the amount but it is still significant that is why usability should still be prioritized.

5. Maintainability

Maintainability is selected as the fifth most important quality attribute in Taobao. According to McCall's Factor Model, maintainability refers to the ease with which a software system can be modified to fix defects, add new features, or adapt to changing environments as the system evolves. For Taobao, this means the platform must be designed so that developers can safely update the search algorithm, recommendation engine, payment integration, and review system without introducing new bugs or causing downtime.

The risks addressed in maintainability are related to the technical and project risks, which are identified across multiple modules, particularly the Recommendation module and Review and Rating module.
For example, the recommendation algorithm was trained on Chinese data and may need continuous updates to adapt to international users with different preferences, such as Western fashion styles, seasonal differences, and product sizing. If the system is difficult to modify, developers may struggle to retrain or update the algorithm effectively. This causes the recommendation feature to provide inaccurate suggestions. In addition, the fake-review detection system must continually evolve as fraud techniques evolve and change constantly. Without maintainability, updating these detection rules or improving the algorithm may introduce new errors or defects that affect the credibility of the review system.

To achieve high maintainability, the development team should implement clear documentation, coding standards, and version control systems to ensure that future developers can easily understand and modify the codebase. Other than that, automated testing, such as regression testing, should be conducted whenever updates are made to ensure that new changes do not cause issues to existing features. Continuous integration and deployment pipelines can also help detect errors early when modifications are introduced.

Maintainability is ranked fifth because it affects the long-term evolution of the system rather than the immediate operational failures. While poor maintainability may slow down development and increase the difficulty of implementing new features, it does not directly cause immediate financial losses like payment failures or system downtime. However, over time, low maintainability can lead to increased technical debt, making the system harder and more expensive to maintain as the platform grows internationally.

From a cost perspective, investing in maintainability through proper software design, documentation, and automated testing is cheaper than attempting to fix poorly structured code later. If the system becomes difficult to maintain, developers will require more time and resources to implement updates or fix defects, which increases the development costs as well as slows it down. By prioritizing maintainability early in the development process, long-term maintenance costs can be reduced and the platform can keep evolving to meet the changing needs of all users.

Summary Table

Rank	Quality Attribute	Module	Risk Addressed
1.	Integrity	Payment Module.	● **Technical Risk(Payment Module):** The bugs between Taobao and Alipay integration that causes errors in billing and escrow system failure. ● **Security Risk (Payment Module):** the non compliance with international data laws such as (GDPR and CCPA that can lead to legal consequences, loss of operation license, and data breaches that expose users sensitive information.
2.	Reliability	Payment Module, Order Tracking Module.	● **Operational Risk(Payment Module):** system experience downtime during peak hours causes immediate revenue loss and users abandoning purchase. ● **Technical Risk(Order Tracking Module):** Multiple courier APIs with different standards that cause tracking errors and reduce the users trust. ● **Programmatics Risk (Order Tracking Module):** poor error handling during the external partners failure/outages by showing blank screen instead of helpful messages increases the users frustration.
3.	Correctness	Product Search Module, Recommendation Module, Order Tracking Module, Review & Rating Module.	● **Technical and Business Risks (Product Search Module):** cross-language bugs that cause mismatched listings and inaccurate search results are leading to loss of sales. ● **Business, Technical, & Project Risks (Recommendation Module):** displaying wrong recommendations that reduces the sellers ROI, algorithm of international users is flawed, skipped validation and the lack of cultural SME. ● **Business & Technical Risk (Review & Rating Module):** unable to detect fake reviews that destroys integrity, incorrect average rating calculations, english text detection failures (False positive/negatives)
4.	Usability	Product Search Module.	● **Operational Risk (Product search Module):** poor multilingual navigation and

			unclear filters causes the users to struggle which leads to increased customer support labor cost.
5.	Maintainability	Order Tracking Module, Review & Rating Module.	● **Technical & Project Risk (Recommendation & Review Module):** the difficulty in updating the chinese-trained algorithm for the international users preferences, inability to evolve the fake review detection as the fraud techniques changes.

1.4 Software Quality Management Process

The process that is chosen for the system is the **Quality Assurance Process.** The Quality Assurance (QA) process refers to a Plan-Do-Check-Act (PDCA) model that is used to ensure that software development activities follow defined standards and procedures to prevent defects before they occur.

Plan Stage

In the **Plan stage,** a quality assurance manager will determine clear goals to produce high-quality products and suggest suitable processes to achieve these goals. This involves identifying quality standards, allocating resources, and defining roles and responsibilities. At this stage, the organization can predict any potential problems and come up with strategies before development.

For Taobao, the Plan stage is needed to establish quality objectives to prevent the critical risks that were identified previously. This includes establishing security protocols for the Payment module to prevent escrow system failures as well as avoid non-compliance with international data laws (GDPR, CCPA). In the Product Search module, the Plan stage helps setting up requirements for cross-language translation accuracy to eliminate mismatches between English keywords and Chinese product listings.. Furthermore, the Plan stage creates guidelines for the Review and Rating module to detect and filter reviews that lead to misinformation and prevent false positives and negatives review before deployment.

Do Stage

In the **Do stage**, processes identified in the previous phase are implemented. The organization carries out and executes its quality plan, which includes establishing procedures, training staff, implementing quality controls, and developing the system according to defined standards. This stage transforms plans into action.

In the Do stage, the planned processes are implemented. For the Payment module, this includes encryption protocols and access controls for the Alipay integration to protect financial transactions from unauthorized users. For the Product Search module, building a load-balanced search infrastructure for the search feature and implementing a translation layer with thorough error handling to prevent cross-language mismatches between English keywords and Chinese product listings.. For Order Tracking, the Do stage includes the development of reliable APIs, retry mechanisms, and fallback messages for integrating with multiple couriers, such as Cainiao, FedEx, etc. For the Recommendation module, the algorithm is retrained using international user data (including Western fashion, sizing, and seasonal preferences) to avoid flawed logic that would cause wrong recommendations.For the Review and Rating module the automated fraud detection systems are built and trained on English text to prevent false positives and false negatives, while also flag bot-generated fake reviews before they influence buyers.

Finally, The development teams are trained on secure coding and preventive quality practices, and all code changes are documented for traceability to ensure future vulnerabilities are prevented.

Check Stage

In the **Check stage**, the results of testing and implementation are checked and compared to what was expected in the Plan stage. This helps to see if the products meet the required standards. If the products meet the standards, then the experts move to the final stage. But if they do not meet it, the team returns to the Plan stage to make any necessary improvements.

For Taobao, the Check stage is essential for validating the quality of implemented features. This includes conducting heavy security checks on the Payment module to identify vulnerabilities before attackers can take advantage of it, as well as performing extensive load testing to ensure there's no downtime during peak international hours. For the Product Search module, running translation accuracy tests using diverse English search terms to confirm there are no mismatches. For the Order Tracking module, the Check stage verifies API integration accuracy with each courier, and simulates partner outages to confirm informative error messages to appear instead of blank screens. In the Recommendation Module, analyzing the relevance of recommendations by measuring the click-through rates to detect any flawed algorithm outputs. For the Review and Rating Module, tests rating calculation logic for correctness as well as reviewing the effectiveness of fraudulent review detection by sampling some reported reviews and gathering user feedback, and validating that both patterns are flagged before influencing buyers.

Act Stage

In the **Act stage**, the organization takes action to improve the quality plan based on the results of the previous stage. This involves making changes to the quality plan, implementing new procedures, and continuing to monitor the quality results.

In the Act stage for Taobao, the platform takes action based on findings from the Check stage. If penetration testing finds any security weaknesses, encryption standards are enhanced in the Payment module. Infrastructure is scaled to more capacity if load tests indicate it's necessary, ensuring smooth shopping during busy events. In the Product Search module, if translation testing uncovers recurring mismatches, then the search translation layer is redesigned to eliminate cross-language bugs. If order tracking shows poor error handling during courier outages, fallback messages and monitoring alerts are improved in Order Tracking module If recommendation relevance remains low for international users in the Recommendation module, the algorithm is retrained with additional cultural data. For the Review and Rating module, if fake reviews bypass detection, fraud detection rules are updated to recognise new bot patterns, and the logic for English text is refined. Lastly, this stage will also focus on improving the interface based on feedback from international users to stay competitive and user-friendly for a global audience.

1.4.1 Justification

Quality Assurance has been identified as the most suitable Software Quality Management process due to its preventive nature rather than a corrective one. For a large and complex platform like Taobao, it is more cost-effective and crucial to prevent defects than to address them post-release. Considering that Integrity and Reliability are prioritized as key quality attributes, Quality Assurance plays a vital role in supporting these attributes by minimizing security risks, reducing system downtime, ensuring transaction accuracy, and maintaining user trust. Other than that, the implementation of a comprehensive Quality Assurance process is essential for ensuring long-term sustainability, customer satisfaction, and competitive advantage for Taobao.

Summary Table

PDCA Stage	Purpose of Stage	Description
Plan	Define quality objectives, standards, roles, and resources before development begins. Identify potential risks and strategies to prevent defects.	Establish security protocols for the Payment module to prevent data breaches, ensure the Search feature can handle high traffic during events like Singles' Day, create moderation guidelines for the Review and Rating module, and allocate usability testing resources for the new English interface.
Do	Implement the planned processes and develop the system according to defined quality standards, including training staff and applying quality controls.	Implement encryption and access controls for payment transactions, develop load-balanced infrastructure for search stability, integrate APIs with logistics partners for order tracking, train developers on secure coding practices, and implement recommendation algorithms and automated fraud detection systems.
Check	Test and evaluate the system to ensure it meets the planned quality standards and objectives. Identify defects or performance issues.	Conduct security testing on the Payment module, perform load testing on the Search feature during peak events like Singles' Day, verify order tracking accuracy, analyze recommendation performance through user interaction metrics, review fraud detection effectiveness, and collect usability feedback on the English interface.
Act	Improve the quality plan and	Strengthen payment security if

	system based on results from the Check stage by implementing corrective and preventive actions.	vulnerabilities are discovered, increase infrastructure capacity if traffic tests show limitations, improve logistics coordination for delivery tracking, refine recommendation algorithms, update fraud detection rules, and enhance the English interface based on international user feedback.
Justification	Explains why Quality Assurance is the most suitable Software Quality Management process.	Quality Assurance focuses on preventing defects rather than fixing them after release, which is critical for a large e-commerce platform like Taobao. It supports key quality attributes such as integrity and reliability by reducing security risks, preventing system failures, ensuring transaction accuracy, and maintaining user trust. This approach also improves long-term sustainability, customer satisfaction, and competitive advantage.

2. Software Quality Metrics

Software quality metrics are measurements applied and used by the project team and quality assurance (QA) team to improve the software operation. In this section, we will apply metrics to the five key quality attributes identified which are integrity, reliability, correctness, usability, and maintainability. For each attribute, we define the transparency which includes key activities, accountability referring to stakeholders responsible, and the relevant quality standards at these 3 levels which are product/system, project, and process. This structured approach will ensure Taobao enhances visibility, responsibility, and overall quality of the software.

2.1 Quality Metrics related to quality attributes

Integrity - RYAN JULIAN RAJESH

RYAN JULIAN RAJESH - INTEGRITY	
Metrics	Criteria
Product/Systems	Transparency ● **Data encryption implementation :** The Taobao system may implement cryptographic algorithms to protect sensitive data. Data encryption implementation must be done to 100% of sensitive data (payment info) transmitted or stored must be encrypted using AES-256 algorithm. Encryption coverage is measured weekly and the target must be 100%. ● **Unauthorized access detection:** This means that we will automate the logging and alerting in case of any attempts by an unauthorized user trying to access parts of the system, such as the payment system.Threshold: alert within 5 seconds of detection. ≥99.9% of attempts must be logged without failure. ● **Penetration testing:** We can apply regular security checks,around 4

	per year to find new weaknesses before they are exploited and used against the system.Each test must cover the Payment module and achieve ≥90% coverage of critical integrity attack vectors Accountability **Information Security Officer** - The information Security officer is in charge of the overall security and integrity of the Taobao system. **Compliance officers** - This is because they are responsible for ensuring that the system is compliant with the international data protection laws. **Quality Assurance Team** - They are responsible for conducting security testing on the system, which includes penetration testing and validation testing of the integrity requirements before the software is released. Quality **ISO/IEC 25010** is the most relevant quality standard for software and system quality. It applies to Taobao's payment module as it provides the system with a technology-independent framework to help measure and ensure the system prevents unauthorized modifications to transaction data and user information. This is specifically important for systems that handle millions of users daily and their transactions as well. We are able to reduce the chances of data breaches.
Project	Transparency

	● **Conducting code reviews and peer inspection processes:** These structured walkthroughs are designed to allow the developers to present their code to peers. Code reviews and peer inspection 100% of code changes to Payment, Order Tracking, and Review modules must undergo peer review. ≥2 reviewers per change. ● **Security testing**: We must apply static and dynamic testing to verify the compliance with security guidelines when developing the system. The dynamic analysis must be run weekly and ≥95% of critical security rules must pass before merge. ● **Defect tracking and resolution reporting**: We should log every integrity-related defect that is found during the development using formal tracking procedures. Critical integrity defects must be fixed within 24 hours; High within 72 hours. 100% traceability from defect to resolution report. Accountability **Project manager** - They are mainly responsible for ensuring the integrity-related requirements are properly defined and implemented during the development. **Software Quality Assurance Group -** They are responsible for ensuring that the project is well documented and nothing is forgotten and unimplemented related to the project's integrity. **Security architect** - They must design the security features into the shape of the system; this includes adding a layer of authentication between the database and the system.

	Quality **ISO/IEC 25023** standard is the most relevant as it describes how we can apply the quality characteristics to measure the project's integrity. The standard specifically provides steps to identify and eliminate structural weaknesses related to integrity before they cause vulnerability problems. The standard is a set of practices that were designed for every phase of the software development cycle to integrate security.
Process	Transparency ● **Planning security protocol definitions:** This means we must creat transparent documentation that contains the security protocols, encryption standards, and even access control requirements for high security modules such as the payment module. 100% of modules (Payment, Search, Recommendation, Order Tracking, Review) must have documented integrity requirements reviewed by Security Architect. ● **Conducting incident response and root cause analysis:** This means we must document any integrity failures and their root cause as well as the corrective actions implemented to solve the issues. The root cause analysis report must be produced within 5 business days and all 100% of corrective actions tracked in a log with closure dates. ● **Conducting regular security audits**: This involves scheduling independent assessments of the effectiveness of the implemented security features and measures in the system. The auditors must verify ≥99% compliance with defined integrity controls.

	Accountability **Development Engineers** - They are responsible for conducting the unit testing for their own modules making sure that all paths through a software routine. These include error paths that should be covered. **Security testing team -** They are responsible for conducting static testing during planning and dynamic testing after deployment and also perform penetration testing to find vulnerabilities earlier and fix and correct them before it is detected by unauthorized personnel's **Auditors** - They must design the security features into the shape of the system; this includes adding a layer of authentication between the database and the system. Quality **ISO/IEC 25010** as applied directly to the quality management process. It provides measures for evaluating allowing us to mitigate the potential of risks to the system process. As an example for the payment module we are able to use threat modelling to ensure that the security logic cannot be bypassed.

Reliability - NATALIE KOA HAO YEE

NATALIE KOA HAO YEE - RELIABILITY	
Metrics	Criteria
Product/Systems	Transparency ● **Reliability Engineering and Design:** Implement Design for Reliability (DfR) and redundancy planning such as failover systems. This will be implemented by using a reliable design,simplifying complex integrations(limit to ≤5 third party APIs per module) , and setting up at least 2 backup servers per region that are configured to take over immediately if the main system fails, the servers will be tested weekly and must have an outcome of 100% success rate) ● **Testing:** schedule load testing before peak events. Implemented by simulating traffic that is 2 to 3 times the amount of normal volume (e.g., 2 million requests/minute), that is performed 4 times per year, with a target of ≤0.5% error rate during the test. This ensures the system is stable during important events. ● **Monitoring:** monitoring latency, traffic, errors and saturation and implementing an automated alert system. This will be implemented by setting up observability of the four signals every 60 seconds and configuring alerts to notify the team if it falls below baseline for ≥2 consecutive minutes. (e.g., latency > 300 ms, error rate > 0.1%) ● **Analysis and Optimization:** Root Cause Analysis (RCA) for reviewing post failures root causes and implementing fixes. Implemented by investigating the cause of why reliability fails and identifying the root cause, with an RCA that is to be completed within 5 business days for all P1/P2 failures, and ≥2 corrective actions are documented per RCA to prevent failures from reoccurring.

	Accountability ● **Site Reliability Engineer (SRE)** - responsible for designing, implementing, and managing system infrastructure. Such as maintaining the target uptime of 99.9%, reported monthly. ● **Technical Team** - Monitors the "Four Golden Signals" and responds to the alerts. For example, they will respond to 100% of P1 alerts and acknowledge them within 5 minutes. ● **Development Team** - implements error handling and performs RCA to fix root causes within 5 days. ● **Quality Assurance (QA) Team** - performs load testing to verify system stability before peak events. Quality The **ISO/IEC 25010** will be used to measure Taobao systems reliability through these four sub-characters: ● **Maturity** - the system is free from failures during normal operation. ● **Availability** - the system is available at all times and backup failover is prepared. ● **Fault tolerance** - such as displaying informative messages when external APIs fail. ● **Recoverability** - the system recovers quickly from failure and RCA is performed to prevent failure from reoccurring.
Project	Transparency ● **Planning and Goal:** Define reliability improvement goals such as

	increasing uptime for the system. This will be implemented by creating measurable targets, such as increasing the uptime from 99.9% to 99.99% within 3 months, with the progress that will be reviewed weekly (4 status reports per month) and a 100% of the goals will be documented in a project charter. ● **Design and Risk Analysis:** Failure Mode and Effects Analysis (FMEA) will be used to identify any potential failure points in the whole system. This will be implemented by systematically analyzing the potential risk of each component (minimum 5 components such as payment, order tracking, search, recommendation, and review) and designing the appropriate redundancy plan, with ≥3 failure modes identified per component and 1 FMEA report per component to be completed before the design phase ends. ● **Execution and Verification:** Verify that all components meet the reliability specifications before deploying. Implemented by conducting integration testing across all modules (minimum 200 integration test cases per release) and ensuring proper documentation of procedures for when any component fails before release, the documentation will cover 100% of critical components with ≥5 steps per failure procedure with an ≥98% passing rate required before it's released. ● **Defect Tracking and Reporting:** Log every defect found during the project using formal tracking procedures. This will be implemented by documenting failure reports where 100% of defects will be logged in Jira within an hour of discovery, assigning ownership(1 owner per defect) , and tracking the fix status with weekly defect summary reports and it should be 52 reports generated per year showing the number of open vs closed defects and the average of open defects in days.

	Accountability ● **Project Manager** - Responsible for defining the goals and creating measurable targets. ● **Reliability Engineers** - Responsible for conducting the FMEA to identify the potential failure points across the system. ● **System Architect** - Responsible for systematically analyzing the risks of each component and designing the appropriate redundancy plan. ● **Development Team** - Responsible for verifying the improvements meet specifications and documenting error handling procedures. ● **QA Team** - Responsible for conducting testing and tracking reliability defects. Quality ● **ISO/IEC 25023** standard is relevant and will be implemented as it provides a framework to apply quality characteristics throughout development. The standard provides steps to help identify and fix any structural weaknesses related to reliability before they lead to system failures or downtime. These practices can be used across all phases of the software development cycle to build reliability into the system from the start.
Process	Transparency ● **Incident Management:** Restores service quickly after failures occur. Implemented by logging 100% of failures with the appropriate severity levels (P1-P4), and assigning an engineer to resolve the issue within the target time, such as P1 ≤15 minutes, P2 ≤4 hours, P3 ≤3 days, P4 ≤5 days. Assignment must occur within 5 minutes of

	deception for P1/P2 severity levels. ● **Problem Management:** prevents failures from recurring. Implemented by conducting RCA within 5 business days for 100% of P1 and P2 incidents that happen again, and maintaining a known Error Database with its workarounds documented for ≥90% of recurring errors. The database will be reviewed weekly (52 reviews per year) ● **Availability Management:** Measures and maintains the target system uptime. Implemented by setting 99.99% availability targets, monitoring the "Four Golden Signals" every 60 seconds (1440 measurements per day), and configuring the automated alerts if signal falls below average ≥2 minutes. Alert logs will be reviewed daily (365 reviews per year). ● **Change Enablement:** Ensure the fixes applied do not break anything. Implemented by requiring the Change Advisory Board (CAB) approval for 100% of all production changes, with CAB holding meetings twice per week (104 meetings per year), and assessing the risk before deploying where the risk assessment is to be documented for every change with a ≥3 success criteria. Change success rate is targeted at ≥95% (no incident within 48 hours). ● **Continual Improvement:** Improve continuously. Implemented by reviewing the Mean Time to Detect (MTTD) and Mean Time to Repair/Respond (MTTR) trends monthly (12 reviews per year), and using lessons learned back into the process, with an improvement actions of ≥2 that is identified per review and 90% completion rate within 3 months. Accountability ● **Incident Manager** - owns the incident process and ensures tickets are logged properly.

	• **On Call Engineer** - Responds to the P1-P4 incidents and restore service within set time. • **Problem Manager** - Leads the RCA investigations and maintains the Known Error Database. • **SRE Team** - sets uptime targets, monitors the four signals, and configures the sales alerts. • **CAB** - Reviews and approves all changes. • **Development Team** - Present the risk assessment and the plan to be implemented for approval. • **QA Manager** - Reviews the MTTD/MTTR trends and feeds the improvement back into the process. Quality • **ISO/IEC 20000-1** measures the effectiveness of the IT reliability processes. Tracks the metrics like the MTTR, RCA completion rate, change success rate, and the availability targets against the defined Service Level Agreement (SLAs)

Correctness - Tiong Chang Vui

TIONG CHANG VUI - CORRECTNESS	
Metrics	Criteria
Product/Systems	Transparency ● **Output Verification Testing:** Automated tests run to verify calculations (e.g., rating averages, payment totals) match expected mathematical results. Implemented by executing a test suite of minimum 500 unit and integration tests per release, with a required pass rate of ≥99.5% . Tests are run daily (365 runs per year) via CI/CD pipeline. ● **Data Consistency Monitoring:** Real-time monitoring ensures UI displayed data (price, stock, delivery date) matches master database records. Implemented by checking data consistency every 60 seconds (1,440 checks per day), with a target match rate of ≥99.9% . Any mismatch triggers an alert within 2 minutes of detection. ● **Cross-Language Semantic Validation:** Testing translation layer to confirm English keywords retrieve correct Chinese product listings. Implemented by sampling 1,000 search queries per week (52,000 queries per year) across 5 international regions, with a target semantic accuracy of ≥98% . Results documented in weekly reports. ● **Algorithm Accuracy Auditing:** Regular sampling of Recommendation Engine output to verify suggestions based on actual user behavior. Implemented by auditing 500 random user sessions bi-weekly (26 audits per year), with a target precision rate of ≥95% alignment between suggested items and actual user purchase or click behavior. ● **Defect Leakage Rate:** Correctness defects escaping to production tracked monthly. Target of <0.5% of all correctness-related defects

	(e.g., miscalculations, wrong item display) reaching end users per release cycle. Accountability ● **Development Engineers:** Write unit tests achieving minimum 90% code coverage for all calculation logic modules (Payment, Rating, Search). Coverage reported weekly. ● • **Quality Assurance (QA) Team:** Executes functional and regression test suites with a required pass rate of ≥98% before release approval. Test results documented in release readiness reports. ● **Localization Specialist / Product Manager:** Validates cross-cultural and multilingual outputs with an error rate target of <2% in sampled queries (200 queries per language per week). ● **Database Administrator (DBA):** Ensures data integrity constraints enforced with zero tolerance for "out of stock" items displayed as "available." Daily integrity reports reviewed. Quality Standard ● **ISO/IEC 25010 (Functional Suitability – Functional Correctness)** Offers a framework to measure the accuracy of a product or system when used by intended users. For Taobao, this ensures mathematical precision of payment calculations, semantic accuracy of search results, and truthful representation of inventory and seller ratings.
Project	Transparency ● **Requirement Validation with SME:** Structured walkthroughs with

	Subject Matter Experts (SMEs) in target international markets to ensure "correct" behavior aligns with local expectations before coding. Implemented by conducting minimum 1 validation session per target region (e.g., 5 regions = 5 sessions) with 100% sign-off on correctness requirements before development begins. Progress tracked via project charter reviewed weekly (4 updates per month). ● **Integration Testing Protocol:** Formal testing to verify data is not corrupted when transmitted between Taobao frontend and backend systems (e.g., UI to Alipay API). Implemented by executing minimum 150 integration test cases per release, with a required pass rate of 100% before deployment approval. Test results documented in release reports. ● **Defect Tracking and Classification:** Logging all "Correctness" defects in Jira for root cause analysis. Implemented by ensuring 100% of correctness defects are logged within 1 hour of discovery, assigned to 1 owner per defect, and tracked with weekly defect summary reports (52 reports per year) showing open vs. closed counts and average age of open defects in days. ● **Logic-Based Review Checks:** Peer reviews focused on fake review detection algorithms to reduce false positives/negatives. Implemented by requiring 100% of algorithm logic changes to undergo peer review with minimum 2 reviewers per change, documented approval before merge. Accountability ● **Project Manager:** Ensures schedule allocates time for SME consultation and requirement validation. Tracks 100% completion of validation sessions and sign-offs. ● **System Architect:** Designs interfaces between modules to prevent data loss or corruption. Verifies integration test pass rate of 100%

	before release. ● **QA Lead:** Monitors density of "Correctness" defects, targeting <0.2 defects per 1,000 lines of code (KLOC). Publishes a weekly dashboard to stakeholders. ● **Business Analyst:** Documents clear acceptance criteria defining "correct" output, with 100% traceability from requirements to test cases. Quality Standard ● **ISO/IEC 25023** This quality standard provides specific measures for evaluating Functional Correctness during the project lifecycle. It guides quantification of accuracy (e.g., percentage of accurate search results) and tracking improvements over time.
Process	Transparency ● **Fake Review Detection Tuning:** Weekly evaluation of bot-detection algorithm accuracy by analyzing false positive and false negative instances. Implemented by sampling 200 flagged reviews per week (10,400 reviews per year), with targets of <5% false positive rate and <3% false negative rate . Weekly accuracy reports generated (52 reports per year). ● **Algorithm Retraining Lifecycle (SDLC Maintenance):** Defined process for retraining Recommendation Engine with updated international user data. Implemented by retraining every 14 days (26 retraining cycles per year) using data from minimum 50,000 international user sessions per cycle. Improvement measured by recommendation relevance score increase (e.g., from 82% to 91% accuracy).

	● **Regression Testing Procedure:** Mandatory execution of "Correctness Test Suite" on code changes to Payment, Search Translation, or Rating modules. Implemented by running 150+ automated regression test cases with a required pass rate of 100% before any change is deployed. Tests run on every commit (estimated 500+ runs per year). ● **Inventory Synchronization Audit:** Daily automated comparison of online stock display vs. warehouse inventory. Implemented by running a reconciliation script once per day (365 audits per year) with a target discrepancy rate of <0.1% across all product listings. Any mismatch triggers an alert to Operations Manager for resolution within 4 hours . ● **Root Cause Analysis (RCA):** For correctness defects that escape to production. Implemented by completing RCA within 3 business days for 100% of P1/P2 correctness incidents, with minimum 2 corrective actions documented per RCA and tracked to 90% completion within 3 months . ● **Process Documentation Updates:** QA Manual reviewed and updated quarterly (4 updates per year) with documented change log of all correctness process modifications. Accountability ● **Process Owner (QA Manager):** Maintains documentation for Regression Testing and Algorithm Tuning processes. Publishes weekly detection accuracy reports and monthly regression pass rate summaries. ● **Lead Data Scientist:** Oversees retraining lifecycle of recommendation and fraud detection models. Documents retraining cycles with before/after accuracy improvement metrics. ● **Operations Manager:** Ensures daily inventory audit runs

	successfully; resolves discrepancies within 4 hours of detection. ● **Change Advisory Board (CAB):** Approves changes to critical correctness-related code after verifying 100% regression test pass evidence. CAB meets twice per week (104 meetings per year). Quality Standard ● **ISO/IEC 20246 (Software and Systems Engineering – Work Product Reviews)** This quality standard ensures systematic inspection of correctness work products with measurable review coverage and defect detection rates.

Usability - WONG KAI LIT KERRY

WONG KAI LIT KERRY - USABILITY	
Metrics	Criteria
Product/System	Transparency ● **Filter and Sort Function Tracking:** Track the success rate of click on filter (price, brand, location) and sort (popularity, sales) functions to test the intuitiveness of the interface. Measured daily via front-end event logs. Measurement: 95% success per button (365 measurements/year). ● **Multilingual Consistency Checking:** Check consistency multilingual interface (Chinese/English) display consistency to show translation mistakes, missing buttons, or layout breakage. Target: ≥ 99% consistency (≤1 error per 100 pages, 365 checks/year). ● **Task Completion Measurement:** Automatically measure the average of steps to finalize a purchase activity (searching, filtering, checkout) to determine unnecessary workflows. Recording of sessions on 100% of transactions. Goal: less than 5 steps per purchase; weekly reports (52

	weekly reports per year). ● **Support Ticket Analysis:** Number of customer support tickets due to confusing UI elements such as hidden cart or payment button. Monthly reviewed and tagged tickets. Goal: 5 percent of all tickets are usability-related, and this figure is cut by 40 percent in 3 months; monthly report (12 reports/year). Accountability ● **UI/UX Designer** - creates clear navigation, filters, and multilingual design. ● **User Researcher** - conducts user testing with international users (non-Chinese users) and gathers up points of confusion. ● **Front-End Developer** - applies auto-correction, adaptive layout and error messages that are helpful. ● **Customer Support Team** - assigns usability-related tickets to give us improvement information. Quality: ● **ISO 9241-11** measured weekly effectiveness (task completion rate of 95% or higher), efficiency (average task time of 60 or less), and satisfaction (user score of 4.5/5).
Project	Transparency ● **User Testing with Real Users:** User test on basic functions (search, filter, view product) of the core functionality with real environment users to document failure steps and points of confusion. 15 international users every quarter, seeing 100% of core activities before every significant release. 4 test sessions per year. ● **A/B Testing of Layouts:** Conduct A/B testing on various filter layouts and button copy and select designs that have a higher task completion rate. 50/50 traffic split for 2 weeks. Goal: 5 or more tests A/B each quarter; must

	not be adopted unless it improves by 5% or more ($p<0.05$). ● **Usability Defect Tracking:** Have usability defect tracking in place to address interface confusion and inconsistent navigation as formal defects with a prioritized fix. 100% of defects registered within 1 hour in JIRA (usability category). P1/P2 fixed ≤3 days, P3/P4 ≤10 days. Weekly defect report (52 reports/year). ● **Cross-Language Auto-Correction Testing:** Conduct cross-language auto-correction testing (English type, Chinese Pinyin) to ensure correctness of suggestion. Weekly test suite of 500 common typos. Target: ≥98% accuracy (52 runs/year). Accountability ● **Project Manager** - makes sure that usability requirements are covered in the project plan and devotes time to user testing. ● **QA Engineer (tester)** - performs usability test cases, records problems, and monitors fixes. ● **Product Manager** - makes decisions on final filter layout and copy that are based on user feedback and data. ● **Localisation Specialist** - ensure that the translation is correct and culturally adapted (date format, currency symbols). Quality ● **ISO/IEC 25023** measured: task completion time (≤ 60s), error rate(less than 5 percent), user satisfaction(at least 4.5/5).
Process	Transparency ● **Regular Usability Audit Process:** Every quarter, an independent team walks through Taobao international version, produces problem lists and

	improvement suggestions. 4 audits per year, covering 100% of core user journeys (search, filter, checkout). Report due in 5 business days with ≥10 actionable recommendations. ● **Root Cause Analysis (RCA) Process:** When "can't find function" tickets spike, mandatory analysis to determine if cause is design, documentation, or training. Trigger RCA within 3 business days of a 50% ticket increase (30-day rolling average). Complete RCA in 5 business days with ≥2 corrective actions; track to 90% completion within 3 months. ● **Mandatory Pre-Release Usability Testing:** Any change to navigation, filters, or payment buttons must pass test with at least 5 international users before release. 100% of UI changes tested; failure rate <10% (≥90% pass first time). Monthly compliance report (12 reports/year). ● **User Feedback Loop Process:** Collect app store reviews and support tickets containing keywords like "hard to use" or "can't find", regularly review and assign improvements. Monthly review meeting. Target: 100% of relevant feedback reviewed within 5 business days; top 5 recurring issues assigned to developer within 10 days. Monthly closure report. Accountability ● **QA Manager** - oversees usability audit execution and ensures audit results are used for improvement. ● **Process Owner** - maintains and updates usability testing process documentation. ● **Compliance Officer** - verifies usability process complies with ISO 9241-11 and keeps audit records. ● **Development Team Leader** - ensures front-end developers follow usability coding guidelines (accessibility labels, keyboard navigation). Quality ● **ISO/IEC 25010** (Process assessment part) – Evaluates whether the organisation systematically embeds usability: 4 audits/year, RCA within 3

	days of spike, 100% UI changes tested pre-release, monthly feedback reviews.

Maintainability - LESTER LIAU JIUN LANG

<table>
<tr><td colspan="2">LESTER LIAU JIUN LANG - MAINTAINABILITY</td></tr>
<tr><td>Metrics</td><td>Criteria</td></tr>
<tr><td>Product/Systems</td><td>
<h3>Transparency</h3>

Modular Design and Architecture: Modularity is measured by ≤ 5 modules per subsystem, ≤ 300 lines of code per module, allowing developers to update or fix individual parts without messing up the entire system.
Preventive Maintenance Planning: Preventive maintenance transparency is tracked via 100% of scheduled inspections logged with timestamps, 4 routine maintenance events per quarter, and zero unplanned outages between maintenance cycles..
Documentation and Code Readability: 100% of documentation measured are up-to-date, with a ≥ 90% code readability score, and weekly documentation review completion rate of 100%, reducing the time new developers need to grasp the system.
Automated Testing (TDD and CI/CD): Test transparency is measured by ≥ 85% test coverage (unit + integration), 100% of pull requests triggering automated CI/CD pipelines, and zero regression failures per deployment.

<h3>Accountability</h3>

System Engineers: Responsible for designing for maintainability, implementing modularity, and enabling repair without disrupting other functional parts.
Maintenance Professionals: Crucial in providing feedback on reliability and repairability during design, and managing the day-to-day upkeep and troubleshooting of the system.
Software Engineers: Responsible for continuous refactoring, upgrading, and implementing test cases to ensure long-term, maintainable code.

</td></tr>
</table>

	Quality **ISO/IEC 25010** is the most relevant standard for system level that ensures long-term system viability, encompassing repairability, extensibility, and scalability, ultimately reducing total ownership costs.
Project	Transparency ● **Design for Maintainability (DfM)**: Incorporating maintenance considerations early in the design phase, including modularity (breaking systems into independent components) and standardizing parts to simplify repairs. ● **Accessibility Planning**: Accessibility planning transparency is tracked via 100% of components having documented access time ≤ 15 minutes to ensure that critical components are physically accessible, safe, and easy to reach for inspection or replacement. ● **Documentation and Coding Standards**: Documentation transparency is measured by weekly documentation update frequency (minimum 1 update per module) , 100% of code files containing header comments with author, date, and purpose, and ≤ 2 hours average time for new developer to understand a module. Accountability ● **Maintenance Manager**: Ensures day-to-day operations run smoothly, prioritizes maintenance tasks, and ensures documentation is kept up-to-date. ● **Quality Assurance (QA) Engineer**: Develops automated tests to ensure that changes to the system do not break existing functionality.

	• **Project Manager**: While focused on delivery, they also manage the budget and timeline for maintenance activities, ensuring proper resources are allocated for long-term support. Quality **ISO 25023 Standards:** These models break down maintainability into sub-characteristics: analyzability, changeability, stability, and testability.
Process	Transparency • **Maintainability Planning & Design (Project-Level)**: Measured by reviewing ≥ 3 historical project documents per new project, and achieving ≥ 80% implementation rate of those lessons in current design. • **Implementation & Execution:** Measured by 100% of technicians passing certification (≥ 40 hours) and ≥ 95% of maintenance procedures documented with step-by-step instructions. • **Continuous Improvement & Tracking:** Measured by 100% of maintenance tickets logged with root cause analysis, and ≥ 80% of insights incorporated into docs within 2 weeks. Accountability • **Maintenance Managers**: Responsible for overall strategy, prioritizing tasks, communicating with executives, and ensuring compliance with health and safety standards. They also manage the work execution process and identify bottlenecks. • **Design/Project Engineers**: Integrate maintainability into the design phase by implementing features like modularization, standardization, and

	equipment accessibility. ● **Maintenance Technicians/Engineers**: Perform diagnostic tasks, repairs, and preventative maintenance. They provide feedback based on experience (maintainer experience) to improve design. Quality **IEEE 14764** provides guidelines for the maintenance process within the system life cycle. Key aspects include: **Maintenance Types**: Corrective, adaptive, perfective, and preventive maintenance. **Process Activities**: Implementation, analysis of problems/modifications, and review/acceptance. **Focus**: Ensures that maintenance is planned and monitored to maintain product quality.

2.2 Quality Control Measurement Tools

Quality control measurement tools are techniques used to help identify, analyze, and solve any quality problems in the system. This section will present 5 issues encountered by Taobao, each will be addressed by a team member using the appropriate QC tool. For each issue, the selected tool will be justified accordingly as the objective is to demonstrate how the application of these tools can help enhance the software's quality and improve users satisfaction.

2.2.1 RYAN JULIAN RAJESH

Issue: Taobao's payment module experiences unauthorized access attempts that could lead to fraudulent transactions or data tampering. The security team wants to identify the root causes of these integrity breaches.

QC Tool Selected: Cause-and-Effect Diagram (Fishbone Diagram)

1. **Encourages team collaboration:** By having the Cause-and-Effect diagram we can involve multiple personnels to uncover hidden integrity weaknesses in the payment module. These personnels include the security analysts, developers, QA testers, and operation staff, bringing diverse perspectives that can be helpful.
2. **Identifies root causes:** The diagram does more than just define symptoms as it also systematically tracks back the issue to their own causes. The causes are categorised as People, Process, Technology and Environment.
3. **Visualises relationships:** It allows us to provide a clear visual map that shows how multiple causes can come together and contribute to the integrity failure. This allows easier understanding and makes it easier for the team to prioritise.
4. **Prevents recurrence:** The team can implement permanent corrective actions rather than repeatedly fixing symptoms by identifying the root cause of the issue.
5. **Supports data-driven decisions:** The diagram can be filled with real data from the incident. This allows taobao to allocate resources to the most critical integrity gaps first.

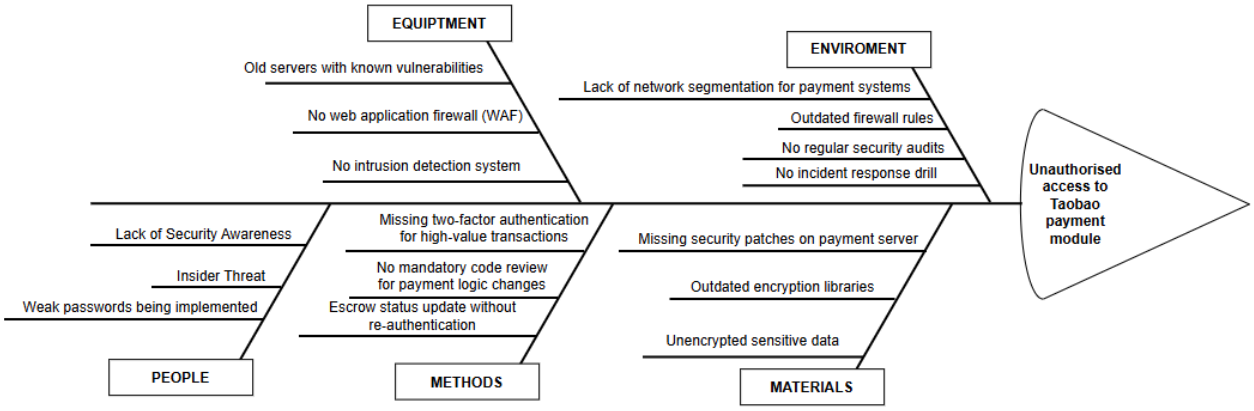

EQUIPTMENT
Old servers with known vulnerabilities
No web application firewall (WAF)
No intrusion detection system
ENVIROMENT
Lack of network segmentation for payment systems
Outdated firewall rules
No regular security audits
No incident response drill
Unauthorised access to Taobao payment module
Lack of Security Awareness
Insider Threat
Weak passwords being implemented
PEOPLE
Missing two-factor authentication for high-value transactions
No mandatory code review for payment logic changes
Escrow status update without re-authentication
METHODS
Missing security patches on payment server
Outdated encryption libraries
Unencrypted sensitive data
MATERIALS

Figure 2.1

2.2.2 NATALIE KOA HAO YEE

issue: Taobao experiences system downtime during peak events, where the traffic spikes overwhelms the system capacity, leading to millions of buyers unable to complete their orders and causing revenue loss. Thus, the company decides to identify which causes to prioritize fixing first.

QC Tool Selected: Pareto Chart

1. **Follows the 80/20 rule:** This helps Taobao identify that 80% of the problem comes from just 20% of the causes, which are responsible for most of the incidents.
2. **Identify priorities for action:** This chart will display which causes have the highest frequency, allowing Taobao to identify the critical issues and fix it first.
3. **Easy to understand for management:** The bar and line chart is simple to understand and read even for non-technical stakeholders like business managers.
4. **Measures improvement over time:** Taobao can compare the Pareto charts before and after fixes to see if the top causes of failures have been reduced.
5. **Prevents wasted resources:** Focuses on the vital few causes rather than wasting time on small minor problems that have little impact.

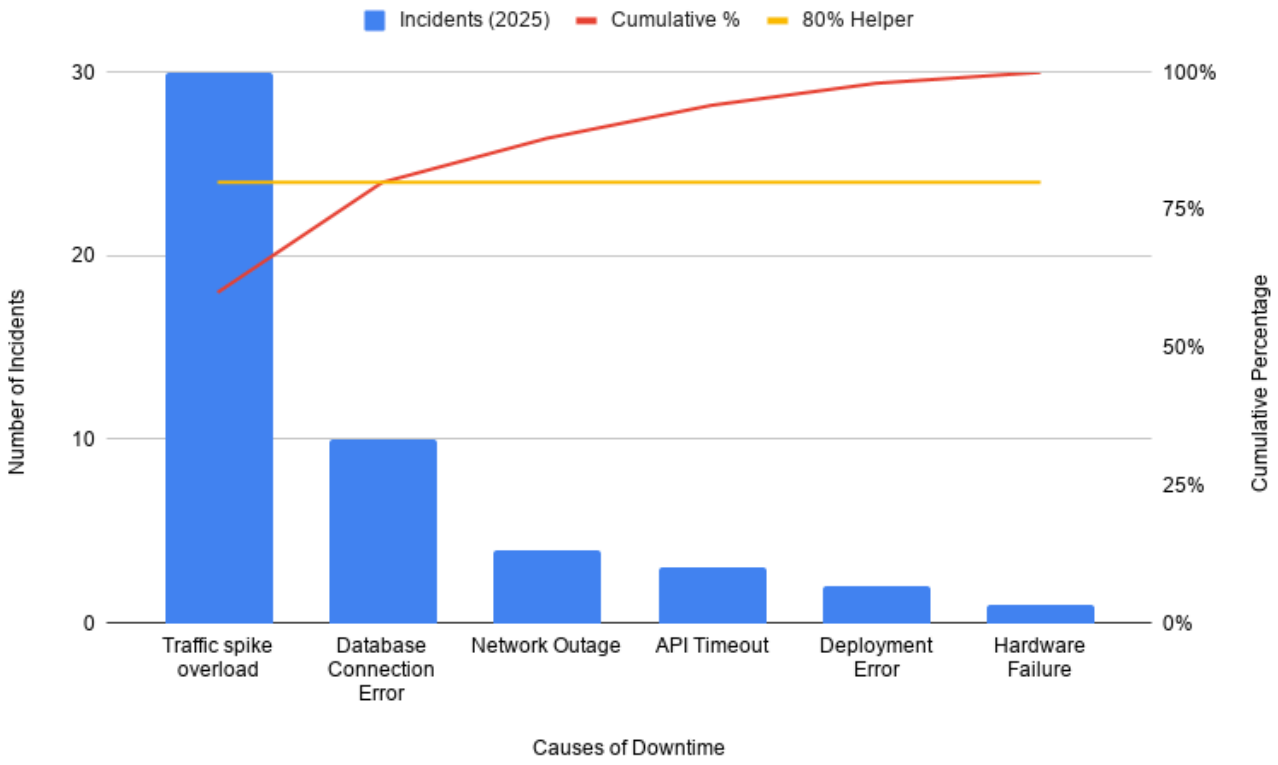

Incidents (2025)
Cumulative %
80% Helper
30
20
10
0
100%
75%
50%
25%
0%
Number of Incidents
Cumulative Percentage
Traffic spike overload
Database Connection Error
Network Outage
API Timeout
Deployment Error
Hardware Failure
Causes of Downtime

Figure 2.2

2.2.3 TIONG CHANG VUI

Issue: The Review & Rating module of Taobao is incorrectly showing the mean star rating of products. For instance, a product receiving reviews with 5 stars and 4 stars would have a mean rating of 2.0 stars. The problem causes buyers to make uninformed decisions and negatively affects the reputation and sales of the seller in question.

QC Tool Selected: Cause-and-Effect Diagram (Fishbone Diagram)

1. **Identifies Root Cause:** Rather than treating the symptoms, this diagram helps dig deeper into the problem and uncover possible root causes such as calculation mistakes or data problems in the DB or cache that help resolve the problem permanently as opposed to addressing the effects of the problem.
2. **Visualizes Complex Connections:** It enables a visual depiction of how two different elements such as the developer's misunderstanding of the business rule and server cache can lead to the same incorrect rating. Such visualization makes it easier for everyone involved to share common ground and communicate effectively between developers/DBAs and business users.
3. **Enables Structured Brainstorming:** The diagram is based on four basic categories which serve as a good starting point for brainstorming. These people (Developers), methods (algorithm logic), materials (QA test data) and machines (Database Administration) should all contribute to a 360-degree view of the problem.
4. **Prevents Repeat of Problems:** Rather than merely fixing the rating for a particular product, the approach helps identify the broken process/code. For example, a problem may occur due to a malfunctioning of the weighted-average function that will have to be fixed so that millions of other products do not get miscalculated.
5. **Improves Data Driven Prioritization:** The diagram can be labeled with real-world facts, such as "Error occurs only when more than 1,000 user ratings". With this information available to the quality control department at Taobao, it is easy to see what causes the problem and work on solving the problem first.

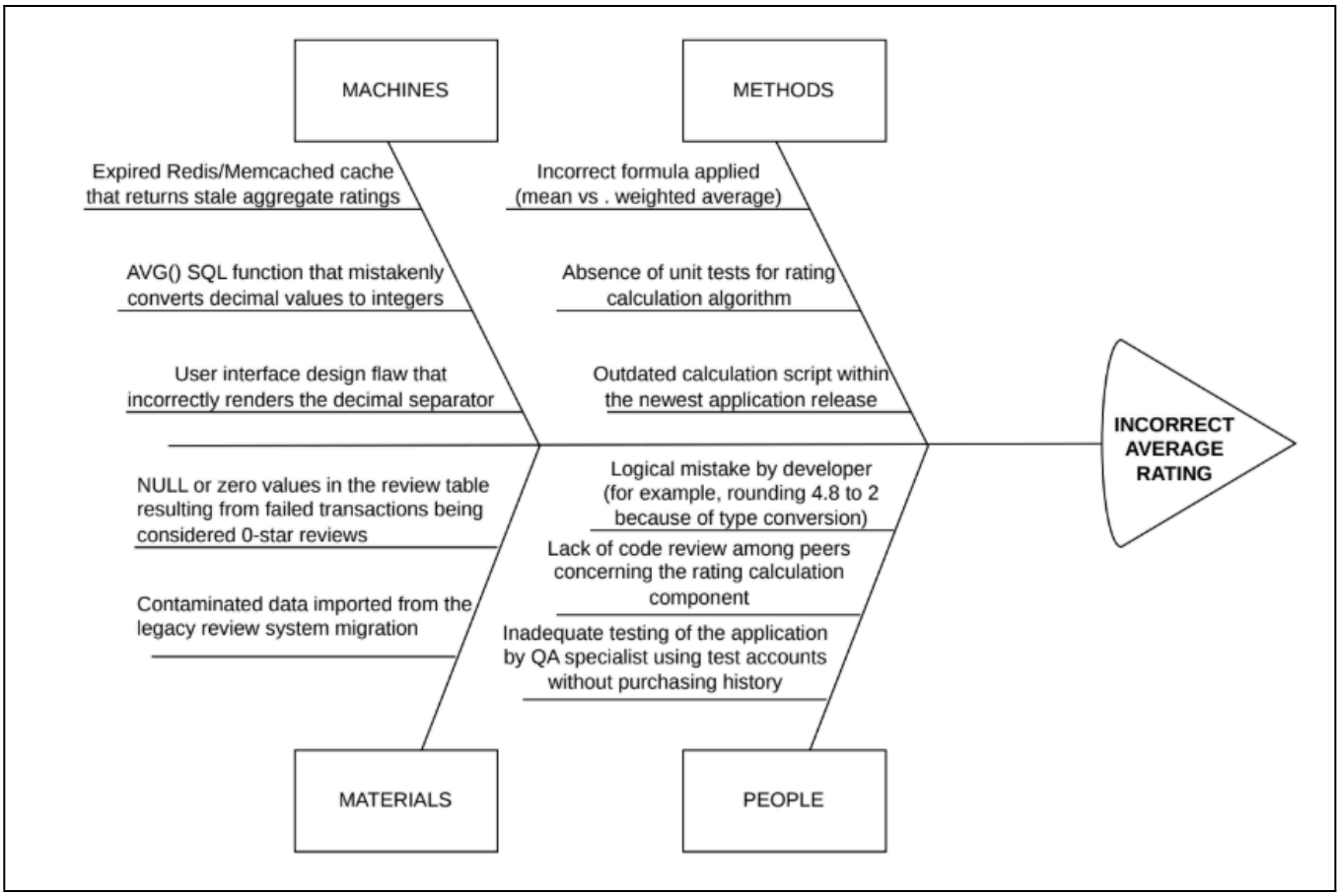

MACHINES
METHODS
Expired Redis/Memcached cache that returns stale aggregate ratings
Incorrect formula applied (mean vs . weighted average)
AVG() SQL function that mistakenly converts decimal values to integers
Absence of unit tests for rating calculation algorithm
User interface design flaw that incorrectly renders the decimal separator
Outdated calculation script within the newest application release
INCORRECT AVERAGE RATING
NULL or zero values in the review table resulting from failed transactions being considered 0-star reviews
Logical mistake by developer (for example, rounding 4.8 to 2 because of type conversion)
Lack of code review among peers concerning the rating calculation component
Contaminated data imported from the legacy review system migration
Inadequate testing of the application by QA specialist using test accounts without purchasing history
MATERIALS
PEOPLE

Figure 2.3

2.2.4 WONG KAI LIT KERRY

Issue: Taobao's international users frequently abandon carts because they cannot find the "Confirm Payment" button or get confused by the checkout workflow. Customer support tickets for "where is the pay button?" have increased by 40% in the last quarter.

QC Tool Selected: Cause-and-Effect Diagram (Fishbone Diagram)

1. **Identifies root causes:** The diagram moves beyond the symptom (cart abandonment) to uncover why users cannot find the payment button, such as poor button placement, misleading labels, or translation errors.
2. **Encourage team collaboration:** It brings together UI/UX designers, front-end developers, customer support agents, and localization specialists to share different perspectives on the usability problem.
3. **Visualize relationship:** The fishbone clearly maps how multiple causes like design, language, process, people, and technology combine to create checkout confusion making it easier to prioritize fixes.
4. **Prevents recurrence:** By identifying the true root cause for example like missing English label on the button, the team can implement permanent corrective action instead of repeatedly answering support tickets.
5. **Support data-driven decisions:** The diagram can be filled with real data from support ticket categories, user session recordings, and A/B test results, helping Taobao allocate resources to the most critical usability gaps first.

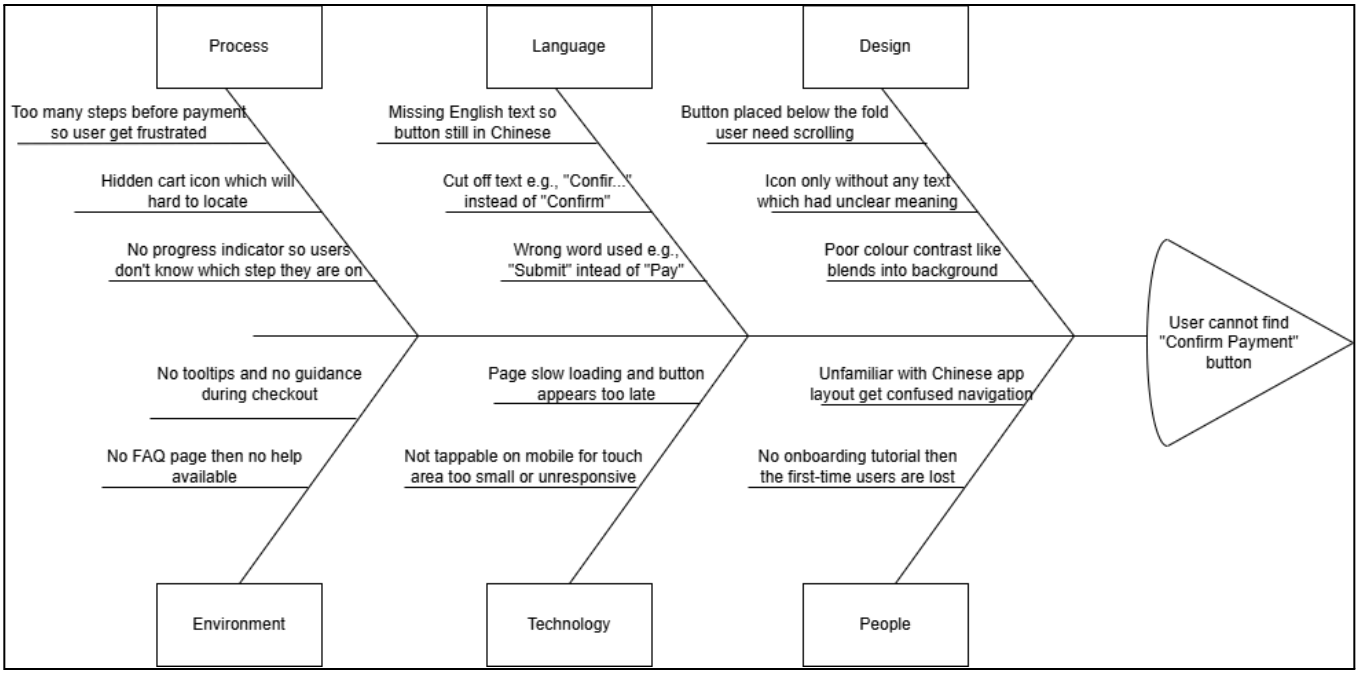

Process
Language
Design
Too many steps before payment so user get frustrated
Hidden cart icon which will hard to locate
No progress indicator so users don't know which step they are on
Missing English text so button still in Chinese
Cut off text e.g., "Confir..." instead of "Confirm"
Wrong word used e.g., "Submit" intead of "Pay"
Button placed below the fold user need scrolling
Icon only without any text which had unclear meaning
Poor colour contrast like blends into background
User cannot find "Confirm Payment" button
No tooltips and no guidance during checkout
No FAQ page then no help available
Page slow loading and button appears too late
Not tappable on mobile for touch area too small or unresponsive
Unfamiliar with Chinese app layout get confused navigation
No onboarding tutorial then the first-time users are lost
Environment
Technology
People

Figure 2.4

2.2.5 LESTER LIAU JIUN LANG

Issue: Taobao's **Review and Rating module** fails to detect fake reviews written in English. The current detection algorithm causes false positives (real English reviews flagged as fake) and false negatives (fake English reviews not detected). This damages marketplace integrity, reduces buyer trust, and negatively impacts sales.

QC Tool Selected: Check Sheet

1. **Systematic data collection -** The check sheet provides a structured format to manually or automatically record each suspected fake review, including attributes like language (English vs Chinese), reviewer account age, star rating pattern, and keyword flags. This ensures no data is missed.
2. **Identifies patterns and trends -** By tallying occurrences of false positives and false negatives over time, the team can detect whether the problem is concentrated on specific product categories, specific English phrases, or specific reviewer behaviors.
3. **Simple and easy to use** - Moderators and QA testers can quickly fill out the check sheet without special training. This allows fast data collection during routine review moderation, reducing the barrier to reporting issues.
4. **Quantifies the problem** - The check sheet converts subjective observations ("many fake reviews are missed") into objective numbers (e.g., "42 false negatives in 7 days"). This data helps management understand the severity and allocate resources appropriately.
5. **Supports root cause analysis** - The collected data from the check sheet can be directly fed into a Pareto Chart or Cause-and-Effect Diagram (used by other team members) to prioritize which type of detection failure (e.g., keyword mismatch, bot pattern, language model error) should be fixed first.

Category	Tally Marks	Total
False Positives (Real review flagged as fake)	||||\ |||	**8**
- Uncommon English spelling	|||	3
- Legitimate negative sentiment	||||	4
- Non-standard punctuation	|	1
False Negatives (Fake review not detected)	||||\ ||||\ ||	**12**
- Bot-generated copied review	||||\ ||	7
- Competitor fake review	|||	3
- Mixed English-Chinese review	||	2
Total Reviews Sampled	**100**	**20 failures**

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